# MASSFormer: Memory-Augmented Spectral–Spatial Transformer for Hyperspectral Image Classification

Le Sun, *Member, IEEE*, Hang Zhang, Yuhui Zheng, *Member, IEEE*, Zebin Wu, *Senior Member, IEEE*, Zhonglin Ye, and Haixing Zhao

***Abstract*— In recent years, convolutional neural networks (CNNs) have achieved remarkable success in hyperspectral image (HSI) classification tasks, primarily due to their outstanding spatial feature extraction capabilities. However, CNNs struggle to capture the diagnostic spectral information inherent in HSI. In contrast, vision transformers (ViTs) exhibit formidable prowess in handling spectral sequence information and excelling at capturing long-range correlations between pixels and bands. Nevertheless, due to the information loss during propagation, some existing transformer-based classification methods struggle to form sufficient spectral–spatial information mixing. To mitigate these limitations, we propose a memory-augmented spectral–spatial transformer (MASSFormer) for HSI classification. Specifically, MASSFormer incorporates two efficacious modules: the memory tokenizer (MT) and the memory-augmented transformer encoder (MATE). The former serves to transform spectral–spatial features into memory tokens for storing prior knowledge. The latter aims to extend traditional multihead self-attention (MHSA) operations by incorporating these memory tokens, enabling ample information blending while alleviating the potential depth decay in the model and consequently improving the model's classification performance. Extensive experiments conducted on four benchmark datasets demonstrate that the proposed method outperforms state-of-the-art methods. The source code is available at https://github.com/hz63/MASSFormer for the sake of reproducibility.**

***Index Terms*— Hyperspectral image (HSI) classification, memory-augmented, multihead self-attention (MHSA), transformer.**

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## I. Introduction

HYPERSPECTRAL imaging is an important technology in the field of remote sensing. Typically, hyperspectral images (HSIs) consist of dozens or even hundreds of narrow and contiguous spectral bands ranging from the visible to the near-infrared, acquired by airborne imaging spectrometers from the same region of the Earth's surface. Therefore, HSI harbors a wealth of spectral information, enabling precise target identification [1] and detection [2], [3] by leveraging the spectral differences among pixels in the HSI. In recent years, HSI has been widely used in multiple fields, including vegetation monitoring [4], precision agriculture [5], urban planning [6], and mineral exploration [7]. These applications encompass various tasks within the realm of HSI processing, such as denoising [8], [9], detection [10], [11], and classification [12], [13]. Among these tasks, HSI classification emerges as a focal point of research, with its primary objective being the assignment of distinct class labels to each pixel within the image.

During the nascent stages of HSI classification research, the majority of classification techniques predominantly emphasized the spectral characteristics of the imagery. These methods encompassed support vector machines (SVMs) [14], random forests (RFs) [15], principal component analysis (PCA) [16], and logistic regression [17]. Nevertheless, owing to their constrained capacity to capture nuanced image details, these conventional approaches frequently underperformed in terms of classification accuracy. Furthermore, these methods suffered from several additional limitations, including a heavy dependence on prior knowledge, suboptimal representation of image features, and diminished generalization capabilities.

In recent years, the swift progression of deep learning technology has captured the substantial interest of researchers. Deep learning empowers the automatic extraction of more profound image features [18]. Consequently, this technology has delivered notable breakthroughs across a spectrum of computer vision tasks [19], encompassing image classification [20], object detection [21], [22], and image segmentation [23]. Moreover, a plethora of deep learning network architectures have emerged, including deep belief networks (DBNs) [24], recurrent neural networks (RNNs) [25], generative adversarial networks (GANs) [26], convolutional neural networks (CNNs) [27], [28], and graph convolutional networks (GCNs) [29].

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The seamless adaptability of deep networks to learning advanced representations has led researchers to incorporate them into HSI classification. For instance, Chen et al. [30] introduced an innovative framework for feature extraction and image classification based on DBN in the analysis of hyperspectral data, demonstrating its effectiveness. In a groundbreaking initiative, Mou et al. [31] leveraged the RNNs framework for HSI classification. They introduced a novel RNN model that adeptly interprets hyperspectral pixels as sequential data and subsequently derives information categories through network inference. Furthermore, Zhu et al. [32] delved into the feasibility and efficacy of GANs in HSI classification. Specifically, a meticulously crafted 1D-GAN served as a spectral classifier, while a robust 3D-GAN functioned as a spectral–spatial classifier. These generated adversarial samples, in conjunction with actual training samples, underwent fine-tuning of distinct CNNs, resulting in enhanced overall classification performance. In a pioneering effort, Paoletti et al. [33] introduced a novel architecture centered around the spectral capsule network (SCN). This redefined the concept of capsule units, customizing them for the spectral–spatial analysis of remote sensing HSI data. The SCN achieved precise HSI classification while notably simplifying network design complexity. Finally, Jia et al. [34] introduced the concept of the graph-in-graph (GiG) model and its affiliated GiG convolutional network (GiGCN) for HSI classification, adopting a superpixel-oriented perspective.

In addition to the aforementioned methods, CNNs have gained widespread popularity in HSI classification [35] due to their inherent characteristics of local connectivity and weight sharing. These attributes make it exceptionally proficient at extracting both rich spectral and spatial features from HSI, rendering CNNs one of the most favored methods in HSI classification. Zhong et al. [36] devised an end-to-end spectral–spatial residual network (SSRN). SSRN operates directly on the original 3-D hyperspectral cubes, eliminating the need for dimensionality reduction of HSIs. Within this network, spectral and spatial residual blocks continuously acquire discriminative features from the abundant spectral characteristics and spatial context within HSIs. Roy et al. [37] proposed the HybridSN for HSI classification. In this innovative approach, the 3D-CNN plays a pivotal role in extracting spatial–spectral joint features from a stack of spectral bands, followed by 2D-CNN to further refine abstract spatial representations. This method effectively reduces the model's complexity when compared to using 3D-CNN in isolation. Wang et al. [38] proposed a multistage self-guided separation network (MGSNet) for remote sensing scene classification. By incorporating texture and morphological features, it significantly enhanced the distinction between samples with target similarity and background differences while simultaneously minimizing differentiation among intraclass samples. Zhou et al. [39] introduced a novel HSI classification model known as the spectral–spatial pyramid network (SSPN), which combines 3-D CNNs with a feature pyramid structure. SSPN leverages the amalgamation of 3-D convolution and multiscale convolution to extract a wide array of spatial–spectral features, thereby significantly enhancing classification performance. Zhang et al. [40] proposed a spatial–logical aggregation network (SLA-NET) for fine-grained tree species classification. This approach integrated spatial convolution and morphological operations into a unified architecture, providing an effective solution for hierarchical feature extraction. This resulted in reducing information loss during data transmission in deep networks. Furthermore, considering the limitations of single-source data, some methods utilized multisource remote sensing data for joint classification [41], [42], which was proven to enhance classification performance. Wang et al. [43] proposed a multistage information complementary fusion network based on flexible-mixup, which achieved effective interaction and enhancement among multimodal information through multiple stages of feature complementation and fusion.

The growing depth of CNNs resulting from the utilization of 3-D convolutional layers can give rise to challenges such as gradient vanishing and exploding. To tackle these issues, researchers have incorporated depthwise separable convolutions (DSCs) and attention [44] mechanisms into HSI classification. Cui et al. [45] introduced a novel network architecture named LiteDepthwiseNet for HSI classification. LiteDepthwiseNet decomposes standard convolutions into depthwise convolutions and pointwise convolutions, achieving high classification performance with a minimal number of parameters. Sun et al. [46] proposed a spectral–spatial attention network (SSAN) to capture distinctive spectral–spatial features from attention regions within HSI cubes. Each spectral and spatial module in SSAN is composed of a limited number of 3-D convolutions and activation operations, resulting in a reduction of the model's parameter count. Dong et al. [47] proposed a weighted feature fusion approach that combines CNNs and the graph attention network, augmented by attention mechanisms, to enhance model performance while mitigating the computational costs associated with graph neural networks. Wang et al. [48] introduced a large kernel sparse ConvNet (LSCNet) with multifrequency attention (MFA) weighting for remote sensing scene understanding. It utilized two parallel rectangular kernels to approximate a large kernel, achieving a larger receptive field and enhancing the model's discriminative and learning capabilities.

Recently, transformer [49] has become a subject of extensive research. Transformer is a deep learning model initially designed to address sequence-related problems, such as speech recognition and natural language processing. The core of the transformer lies in its self-attention mechanism, allowing it to establish global dependencies within input sequences. This unique capability makes it exceptionally well-suited for handling sequential data. Notably, the emergence of the vision transformer (ViT) [50] has marked a significant breakthrough in advancing state-of-the-art technology for visual tasks. ViT's success underscores the versatility of transformer models, showing that they are not confined to text data and can effectively be applied to diverse data types. HSI is characterized by its rich spectral information, which possesses sequential properties. Consequently, researchers have leveraged transformer models for HSI classification, recognizing their potential to capture and exploit the sequential nature of spectral data.

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Hong et al. [51] rethought HSI classification from the sequential perspective of the transformer. They introduced a novel backbone network named SpectralFormer, capable of learning local spectral sequence information from adjacent spectral bands within HSI. This process generates grouped spectral embeddings and results in superior classification accuracy compared to the classical ViT. Sun et al. [52] proposed a method called spectral–spatial feature tokenization transformer (SSFTT), designed to capture both spectral–spatial features and high-level semantic features. This addresses the limitation faced by CNN-based classification methods in extracting deep semantic features. Zou et al. [53] proposed an innovative local-enhanced spectral–spatial transformer (LESSFormer) method, specifically tailored for HSI classification. LESSFormer transforms HSI into adaptive spectral–spatial token representations, enhancing the discriminative power of these tokens by simultaneously capturing local and remote information. Peng et al. [54] proposed a spatial–spectral transformer with cross attention (CASST). In essence, this method consists of spatial sequence branches and spectral sequence branches, enabling efficient HSI classification through the fusion of features from both branches. Inspired by spectral partitioning techniques that emphasize local representation in the spectral domain, Mei et al. [55] proposed a group-aware hierarchical transformer (GAHT) for HSI classification. GAHT confines the multihead self-attention to local spectral–spatial contexts by introducing novel grouped pixel embedding (GPE) modules. Roy et al. [56] proposed a novel morphological transformer (morphFormer) that implements a learnable spectral and spatial morphology network. This approach incorporates spectral and spatial morphology convolution operations to enhance interactions between structure and shape information. Fang et al. [57] proposed a lightweight transformer known as MAR-LWFormer, which effectively fuses the spectral and spatial features of HSI. MAR-LWFormer addresses the challenge of inadequate modeling of spectral–spatial information by ViT in HSI classification scenarios with meager sampling rates. Apart from the classification approach based solely on HSI data, Zhao et al. [58] proposed a novel dual-branch method. This method comprises a hierarchical CNN and a transformer network, with the objective of amalgamating spectral–spatial features from HSI data and elevation features from light detection and ranging (LiDAR) data to improve the overall joint classification performance.

While the aforementioned deep learning methods have demonstrated impressive performance, the inherent high-dimensional [59] nature of HSI and the limited availability of labeled samples [60] continue to present significant challenges for classification tasks. On the one hand, CNN-based approaches excel in processing spatial information but encounter difficulties in effectively utilizing the sequential spectral information present in HSI. In contrast, transformer-based methods are proficient at capturing long-range correlations between spatial and spectral information and perform well when dealing with spectral sequences. However, they do have limitations when it comes to handling spatial information. On the other hand, in certain transformer-based HSI classification methods, spectral–spatial information often has to traverse several encoders to reach its intended location. This process can lead to progressive information degradation or blurring. To address these challenges, this article proposed a memory-augmented spectral–spatial transformer (MASSFormer) network, which combines the strengths of both CNNs and transformers. Specifically, MASSFormer first employs a shallow CNN module to extract spectral–spatial features from the HSI. Subsequently, a simple and efficient module called the memory tokenizer (MT) encodes spectral–spatial features into memory tokens that store prior knowledge. These memory tokens, alongside the spectral–spatial features, are then utilized as input for the memory-augmented transformer encoder (MATE) to facilitate feature learning. MATE leverages information aggregated over the entire input sequence to update vector representations of the features. Finally, the classification token is fed into the MLP head to obtain pixel labels.

The main contributions of this article are summarized as follows.

1) An MT module is proposed to transform spectral–spatial features extracted by a shallow CNN into memory tokens. These tokens serve as repositories for local spectral–spatial context and salient feature representations. They consistently preserve the original information throughout subsequent stages of feature learning, which can be considered persistent memory.
2) By leveraging the prior knowledge provided by memory tokens, we expanded upon the conventional self-attention mechanism to create an MATE. In contrast to the original TE, the integration of this supplementary information significantly augments the abundance of feature representations in the model. This enables the model to achieve ample information blending and mitigates the problem of depth decay in deep architectures, thereby enhancing the overall classification performance.
3) A MASSFormer network is proposed that combines the advantages of both CNN and transformer architectures. Extensive experiments on four public datasets have been conducted to validate the effectiveness of the proposed method.

The remainder of this article is organized as follows. In Section II, a detailed exposition of the MASSFormer network is provided. Section III presents extensive experimental results and analytical discussions. Section IV concludes with a summary and prospects for future work.

## II. Proposed Method

The MASSFormer architecture is illustrated in Fig. 1. First, PCA is applied to the original HSI to effectively reduce data dimensionality. Subsequently, the reduced-dimensional data are segmented into small 3-D patches, serving as input to the feature extraction module. Following this step, a shallow CNN module is employed to extract joint spectral–spatial features. Then, the MT module is utilized to transform these spectral–spatial features into memory tokens, thereby preserving local context and crucial information. Finally, the spectral–spatial features and memory tokens are collectively

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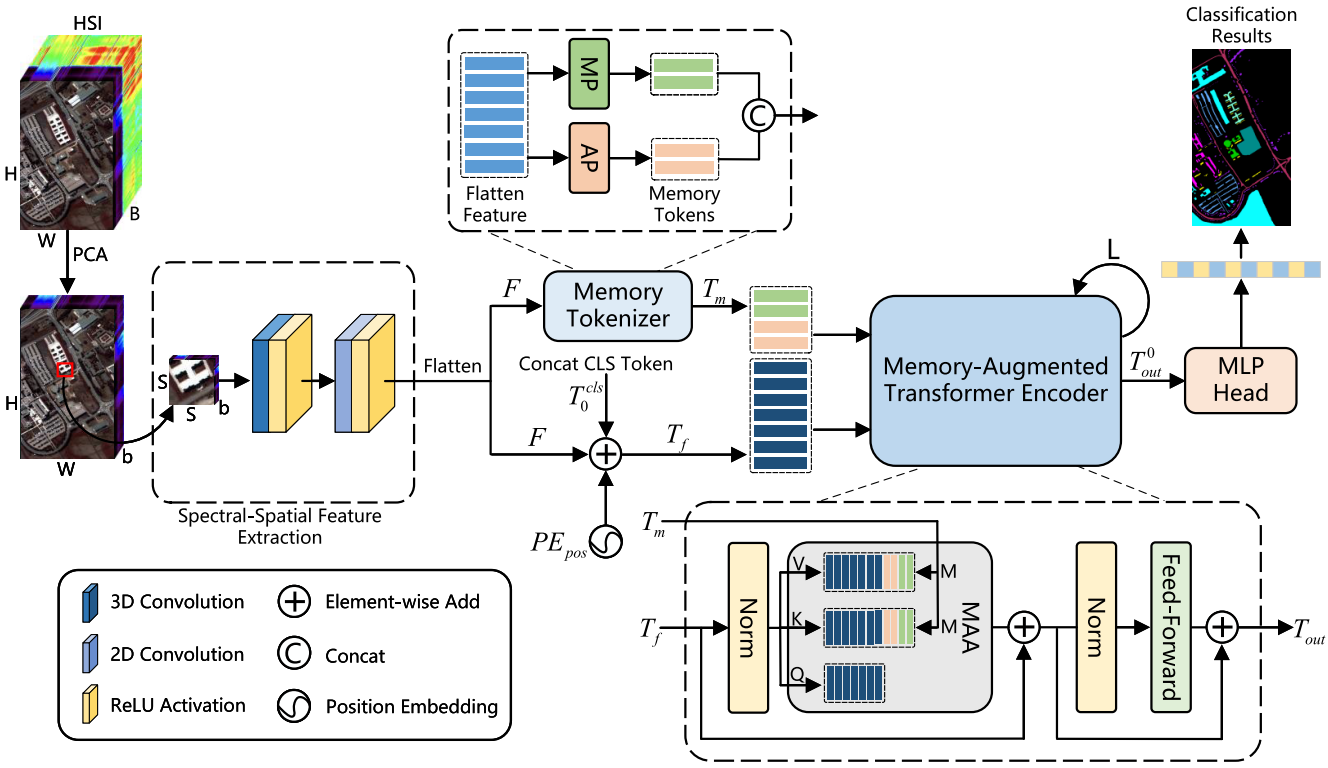



Fig. 1. Overall architecture of MASSFormer for HSI classification.

fed into the MATE for comprehensive feature learning, and the classification token is extracted for the ultimate classification.

### A. HSI Data Preprocessing

Typically, HSIs offer abundant spectral bands that provide valuable information but also introduce a significant amount of redundancy. To mitigate computational complexity, we initially employ PCA to perform dimensionality reduction on the HSI data. Given a hyperspectral data cube denoted as $\boldsymbol{X} \in \mathbb{R}^{W\times H\times B}$, where $\boldsymbol{X}$ serves as the original input to the entire classification framework, and $W$, $H$, and $B$ correspond to the width, height, and the number of spectral bands in the HSI, respectively, the label for each pixel within $\boldsymbol{X}$ is denoted by a one-hot vector $\boldsymbol{Y} \in \mathbb{R}^{1\times 1\times C}$, where $C$ represents the number of land cover categories. Subsequently, PCA is administered to the original HSI data $\boldsymbol{X}$ along the spectral dimension. Following dimensionality reduction, the spatial resolution of $\boldsymbol{X}$ remains unaltered, while the number of spectral bands decreases from $B$ to $b$, i.e., $\boldsymbol{X}_{\text{pca}} \in \mathbb{R}^{W\times H\times b}$. In other words, PCA not only eliminates redundant spectral information in the HSI but also preserves spatial information without degradation.

Subsequently, we divide $\boldsymbol{X}_{\text{pca}}$ into small overlapping 3-D neighboring patches. Each patch is denoted as $\boldsymbol{X}_{\text{patch}} \in \mathbb{R}^{S\times S\times b}$, where $S \times S$ represents the spatial size of the patch and $b$ corresponds to the number of spectral bands. The true label for each patch is determined by the label of the center pixel. For all neighboring patches, take patch $\boldsymbol{X}_{\text{patch}}^{ij}$ whose central pixel position is $(i, j)$ as an example, and its spatial coverage ranges from $i-(S-1)/2$ to $i+(S-1)/2$ in width and from $j-(S-1)/2$ to $j+(S-1)/2$ in height. It encompasses all spectral bands within this spatial extent. It is imperative to highlight that when generating patches for edge pixels, the necessity for padding operations arises due to the asymmetric coverage dimensions, which are less than $(S-1)/2$ on one side of these pixels. Following the generation of all 3-D patches, patches with labels equal to 0 are excluded. The remaining patches are divided into training and testing sets in proportion.

### B. Shallow CNN Module

In recent years, given the exceptional local feature extraction capabilities of CNNs, they have witnessed extensive application in HSI classification. In the proposed MASSFormer, we incorporated a shallow CNN to effectively extract spectral and spatial information from HSI. It is composed of a 3-D convolutional block followed by a 2-D convolutional block. This strategic design facilitates the seamless integration of spectral and spatial information at an early stage of feature extraction.

We commence by introducing a 3-D convolution-based block to process the input 3-D neighborhood patches. This convolution block consists of a 3-D convolution layer with a kernel size of 8@3 × 3 × 3, accompanied by a nonlinear activation layer. The stride and padding employ default values of 1 and 0, respectively. Specifically, the 3-D convolution layer conducts convolution operations along both spectral and spatial dimensions, yielding 3-D feature maps that encapsulate the spectral–spatial joint characteristics. The computation process of the 3-D convolution block is expressed as

$$F_{3\text{d}} = \Phi\left(X_{\text{patch}} \Theta w_{3\text{d}} + b_{3\text{d}}\right) \tag{1}$$

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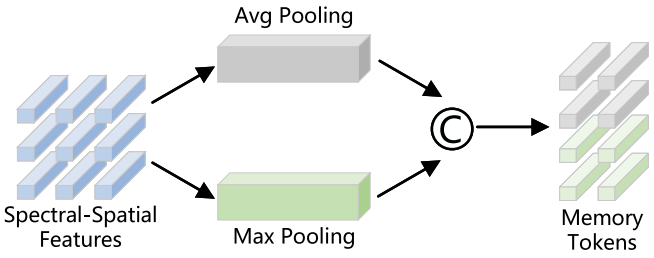



Fig. 2. Illustration of the memory tokenization.

where $F_{3d}$ represents the 3-D feature map, $w_{3d}$ and $b_{3d}$ are the weight and bias parameters, respectively. $\Theta$ is the 3-D convolution operator, and $\Phi$ is the activation function.

Then, following this step, the 3-D feature maps undergo a flattening operation along the spectral dimension, serving as input for the 2-D convolution block. The 2-D convolution block is analogous to the 3-D counterpart and consists of a 2-D convolution layer with a kernel size of 64@3 $\times$ 3 and a subsequent nonlinear activation layer. The 2-D convolution block performs convolution along the spatial dimension to extract more discriminative spatial information

$$F_{2d} = \Phi(f(F_{3d}) \odot w_{2d} + b_{2d}) \tag{2}$$

where $F_{2d}$ represents the 2-D feature map, $f$ is the flatten operation, and $w_{2d}$ and $b_{2d}$ are the weight and bias parameters, respectively. $\odot$ is the 2-D convolution operator, and $\Phi$ is the activation function.

Finally, we flatten the 2-D feature map $F_{2d}$ along the spatial dimension to obtain the final output feature $F$ of the shallow CNN module. With this step, we achieve the exploration of spatial and spectral information within the data at relatively lower computational costs.

### C. Memory Tokenization

Despite the limited representational capacity and relatively scattered distribution of features extracted by the shallow CNN, it contains more fundamental and generic information. This aids the network in capturing the structure and patterns of the input data more effectively, contributing to enhanced generalization capabilities. Therefore, we introduce a straightforward yet effective module referred to as MT; its process is described in Fig. 2. The goal of this module is to store valuable information from the spectral–spatial features persistently, including local contextual representations and salient feature details, thereby augmenting the model's subsequent feature learning capabilities.

Specifically, we commence by executing average pooling on the spectral–spatial features, where the feature values within each pooling window are averaged. This helps in preserving overall patterns and structural information of the feature distribution and better understanding the targets in the image, making samples of the same class more consistent. In addition, we perform max pooling on the spectral–spatial features, selecting the maximum value within each pooling window, emphasizing salient feature points, and suppressing other secondary features. These salient feature points are crucial for distinguishing different categories of samples. From an intuitive standpoint, the result of average pooling is close to but not exactly equal to the values within the window, akin to how humans have vague recollections of ordinary events. The outcome of max pooling matches the maximum value within the window, similar to humans' clear memory for certain significant events. Therefore, we combine the results of these two operations and refer to them collectively as memory tokens, constituting an integrated fusion of overall and detailed information. The generation process of memory tokens can be expressed as

$$T_m = \text{Concat}(\text{AvgPool}(F), \text{MaxPool}(F)) \tag{3}$$

where Concat represents the splicing operation, $F$ represents the spectral–spatial features, and the sizes of AvgPool and MaxPool are 3 $\times$ 1 and 7 $\times$ 1, respectively. $T_m$ can intuitively be regarded as a form of memory for the shallow features and can serve as supplementary information to subsequent modules, thus providing more original feature representations and further enhancing the model's performance.

### D. Memory-Augmented Transformer Encoder

As illustrated in Fig. 1, the MATE primarily consists of a memory-augmented attention (MAA) block and a simple fully connected (FC) feed-forward network (FFN).

The original MHSA mechanism aims to establish a global long-range dependence among the input feature sequence. The input to the MHSA is defined as

$$T_f = \left[T_0^{\text{cls}}, F_1, \ldots, F_n\right] + \text{PE}_{\text{pos}} \tag{4}$$

where $T_0^{\text{cls}}$ is a special learnable classification token, shared across all inputs and used for the final classification prediction; $F_i, i = 1, 2, \ldots, n$, represents the spectral–spatial features; and $\text{PE}_{\text{pos}}$ denotes the positional information appended to the entire sequence.

Now, to provide MHSA with more valuable contextual information, we extend the traditional MHSA mechanism to MAA and take both the spectral–spatial features and memory tokens encoding prior knowledge together as input for MAA, and it can be expressed as

$$T_{\text{in}} = \left[T_f; T_m\right] \tag{5}$$

where $T_m$ represents memory tokens. The MAA module is illustrated in Fig. 3; it first conducts linear transformations on both the feature vectors and memory tokens, resulting in four different matrices, which include query $Q \in \mathbb{R}^{s\times d}$, key $K \in \mathbb{R}^{s\times d}$, value $V \in \mathbb{R}^{s\times d}$, and memory $M \in \mathbb{R}^{l\times d}$; and the linear transformation process is defined as

$$\begin{aligned} Q &= T_f W_q \\ K &= T_f W_k \\ V &= T_f W_v \\ M &= T_m W_m \end{aligned} \tag{6}$$

where $s$ and $l$ represent the number of feature vectors and memory tokens; $d$ denotes their dimensionality; and $W_q$, $W_k$, $W_v$, and $W_m$ are learnable weight matrices. Subsequently, $Q$,

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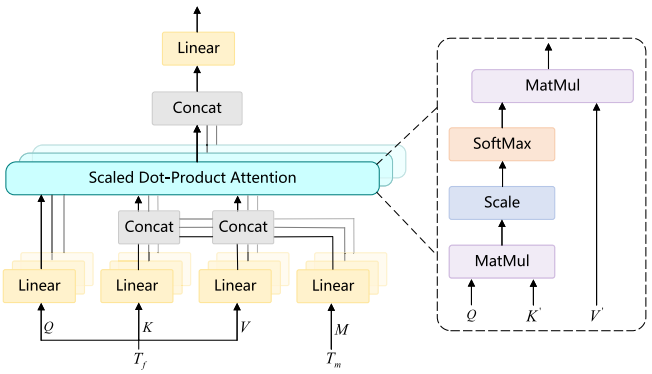



Fig. 3. Illustration of the MAA module.

$K$, $V$, and $M$ are split into $h$ parts along the $d$ dimension, denoted as follows:

$$\begin{aligned} Q &= [Q_1, Q_2, \ldots, Q_h] \\ K &= [K_1, K_2, \ldots, K_h] \\ V &= [V_1, V_2, \ldots, V_h] \\ M &= [M_1, M_2, \ldots, M_h] \end{aligned} \tag{7}$$

where $h$ is the number of attention heads, $Q_i \in \mathbb{R}^{s\times(d/h)}$, $K_i \in \mathbb{R}^{s\times(d/h)}$, $V_i \in \mathbb{R}^{s\times(d/h)}$, and $M_i \in \mathbb{R}^{l\times(d/h)}$. Next, we extend $K_i$ and $V_i$ with $M_i$ as

$$\begin{aligned} K_i' &= \text{Concat}(K_i, M_i) \\ V_i' &= \text{Concat}(V_i, M_i). \end{aligned} \tag{8}$$

Since $M$ represents the projection matrix for memory tokens, it consistently retains local context and salient feature representations. Utilizing $M$ to expand $K$ and $V$ introduces this memory-like component into self-attention operations, enabling the model to consider a broader context during self-attention computations. This aids in providing a more comprehensive information set, facilitating the model in better understanding the relationships and dependencies among different segments within the input sequence. Furthermore, employing multihead mechanisms allows the model to concurrently attend to diverse subspaces of information. The expansion part of $K$ and $V$ for each head offers additional information, augmenting the model's generalization capabilities and improving its predictive performance on unseen data.

Following this, attention scores are calculated for each $Q_i$ and $K_i'$ within every attention head, and these scores are then transformed into attention weights using a softmax function. Finally, these weights are used to perform multiplication with $V_i'$. The computation process for each head is delineated as follows:

$$H_i = \text{Attention}\left(Q_i, K_i', V_i'\right) = \text{Softmax}\left(\frac{Q_i K_i'^T}{\sqrt{d}}\right) V_i'. \tag{9}$$

The final output of the MAA is constructed by concatenating the attention results from all attention heads and further subjecting them to a projection operation. This can be expressed as

$$\text{MAA}(T_{\text{in}}) = \text{Concat}(H_1, H_2, \ldots, H_h) W_o \tag{10}$$

where $W_o \in \mathbb{R}^{d\times d}$ is the parameter matrix.

Then, the output of MAA serves as the input to the FFN. The feed-forward layer comprises two FC layers with a Gaussian error linear unit (GELU) inserted in between; it can be defined as

$$\text{FFN}(X) = \text{FC}_2(\text{GELU}(\text{FC}_1(X))). \tag{11}$$

In conclusion, the entire computational process of MATE can be summarized as follows:

$$z_\ell' = \text{MAA}(\text{LN}(z_\ell)) + z_\ell \tag{12}$$

$$z_{\ell+1}' = \text{FFN}\left(\text{LN}\left(z_\ell'\right)\right) + z_\ell' \tag{13}$$

where LN is the layer norm, which mitigates gradient vanishing and exploding issues, thus accelerating the training process. $z_\ell$ represents the input to the $l$th layer of the MATE. For example, if $l = 1$, it means that $z_\ell$ is composed of $T_f$ and $T_m$. $z_{\ell+1}'$ represents the output of the $l$th layer of the MATE and corresponds to $T_{\text{out}}$ depicted in Fig. 1, where $l$ lies in the range of $[1, L]$ and $L$ is the maximum iterative number for the MATE module.

### E. Classification Head

To achieve the final classification, we employ a multilayer perceptron (MLP) head, MLP typically consists of multiple FC layers, and the MLP head refers to its final layer. In this article, the MLP head comprises a layer norm and an FC layer. The classification token is extracted from the output of MATE and serves as the input to the MLP head, where the output dimension of the MLP head is equal to the total number of classes for the final predictions. The unit with the highest value in this output corresponds to the predicted label for the pixel. The entire execution process of the proposed method is outlined in Algorithm 1.

## III. Experiments and Analysis

In this section, we evaluate the effectiveness of the proposed method through comprehensive experiments conducted on four public datasets. These experiments include comparisons with state-of-the-art approaches. First, a detailed introduction of the four datasets used in the experiments is provided. Subsequently, we delineate various experimental configurations and conduct parameter analyses. Next, we present the classification results and analyses for both the proposed method and comparative methods. Finally, ablation experiments are carried out to demonstrate the effectiveness of individual modules.

### A. Dataset Description

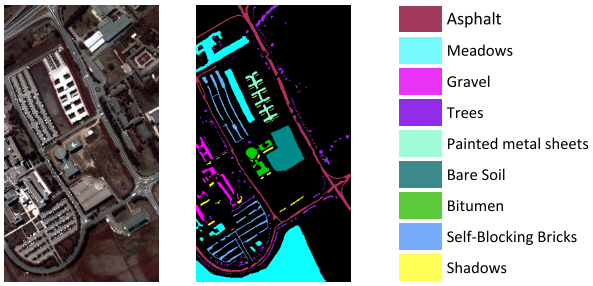
*1) Pavia University:* The Pavia University (PU) dataset was acquired during an aerial survey conducted over Pavia, Northern Italy. It boasts a spatial resolution of 1.3 m and has dimensions of 610 × 340 × 103, where 610 and 340 correspond to the pixel count in the height and width dimensions, respectively, and 103 signifies the number of spectral bands. The PU dataset comprises nine distinct land cover categories, and its pseudocolor composite image along with the ground-truth map is presented in Fig. 4.

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**Algorithm 1** MASSFormer Network for HSI Classification

**Input:** HSI data $\boldsymbol{X} \in \mathbb{R}^{W*H*B}$ and label $\boldsymbol{Y} \in \mathbb{R}^{W*H}$; PCA bands number $b$; patch size $S$; training sample rate $\alpha\%$.
**Output:** Predicted labels of the test set.

```
Obtain the X_pca after PCA transform.
Obtain patches from X_pca and divide them into train set and test set.
Set batch size bs = 32; learning rate lr = 5e-4; epochs e = 100.
for i = 1 to e do
    Perform Conv3D and Conv2D on patches to acquire spatial–spectral features F_3d and F_2d.
    Flatten 2D feature maps F_2d into 1D feature vectors F.
    Generate memory tokens T_m from feature vectors F using MT.
    Obtain T_f by adding positional information and an additional classification token to F.
    Feed the feature T_f and memory tokens T_m into the MATE for feature learning.
    for j = 1 to L do
        Perform MAA operation by Eq. (10).
        Perform FFN operation by Eq. (11).
    end for
    Input the first classification token T_out^0 to the MLP head to obtain the category label.
end for
Employ the trained model to perform predictions on the test set.
```



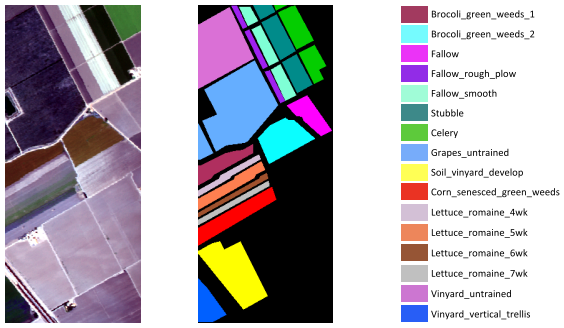

(a) (b)

Fig. 4. PU dataset. (a) False-color map. (b) Ground-truth map.

TABLE I
TRAIN AND TEST SAMPLES IN THE PU DATASET

| ID | Land Cover Class | Train | Test |
|---|---|---|---|
| C01 | Asphalt | 33 | 6598 |
| C02 | Meadows | 93 | 18556 |
| C03 | Gravel | 10 | 2089 |
| C04 | Trees | 15 | 3049 |
| C05 | Painted metal sheets | 7 | 1338 |
| C06 | Bare Soil | 25 | 5004 |
| C07 | Bitumen | 7 | 1323 |
| C08 | Self-Blocking Bricks | 18 | 3664 |
| C09 | Shadows | 5 | 942 |
| - | Total | 213 | 42563 |

*2) Salinas:* The Salinas dataset was collected over the Salinas Valley, CA, United States. It features a spatial resolution of 3.7 m and has dimensions of 512 × 217 × 204, where




(a) (b)

Fig. 5. Salinas dataset. (a) False-color map. (b) Ground-truth map.

TABLE II
TRAIN AND TEST SAMPLES IN THE SALINAS DATASET

| ID | Land Cover Class | Train | Test |
|---|---|---|---|
| C01 | Brocoli_green_weeds_1 | 10 | 1999 |
| C02 | Brocoli_green_weeds_2 | 19 | 3707 |
| C03 | Fallow | 10 | 1966 |
| C04 | Fallow_rough_plow | 7 | 1387 |
| C05 | Fallow_smooth | 13 | 2665 |
| C06 | Stubble | 20 | 3939 |
| C07 | Celery | 18 | 3561 |
| C08 | Grapes_untrained | 56 | 11215 |
| C09 | Soil_vinyard_develop | 31 | 6172 |
| C10 | Corn_senesced_green_weeds | 16 | 3262 |
| C11 | Lettuce_romaine_4wk | 5 | 1063 |
| C12 | Lettuce_romaine_5wk | 10 | 1917 |
| C13 | Lettuce_romaine_6wk | 5 | 911 |
| C14 | Lettuce_romaine_7wk | 5 | 1065 |
| C15 | Vinyard_untrained | 36 | 7232 |
| C16 | Vinyard_vertical_trellis | 9 | 1798 |
| - | Total | 270 | 53859 |

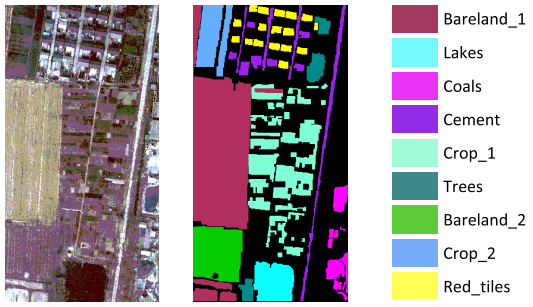
512 and 217 represent the pixel count in the height and width dimensions and 204 denotes the number of spectral bands. The Salinas dataset encompasses a total of 16 distinct land cover categories, and its pseudocolor composite image along with the ground-truth map is displayed in Fig. 5.

*3) Xuzhou:* The Xuzhou dataset was captured over Xuzhou, Jiangsu province, China. It boasts a spatial resolution of 0.73 m and has dimensions of 500 × 260 × 436, with 500 and 260 representing the pixel count in the height and width dimensions and 436 indicating the number of spectral bands. The Xuzhou dataset includes nine distinct land cover categories, and its pseudocolor composite image and ground-truth map are illustrated in Fig. 6.

*4) WHU-Hi-HanChuan:* The WHU-Hi-HanChuan dataset was acquired over Hanchuan, Hubei province, China. It possesses a spatial resolution of 0.109 m and has dimensions of 1217 × 303 × 274, where 1217 and 303 represent the pixel count in the height and width dimensions and 274 denotes the number of spectral bands. The WHU-Hi-HanChuan dataset comprises a total of 16 distinct land cover categories, and its pseudocolor composite image and ground-truth map are depicted in Fig. 7.

The land cover class names, as well as the numbers of training and testing samples used in the experiments for the mentioned four datasets, are presented in Tables I–IV.

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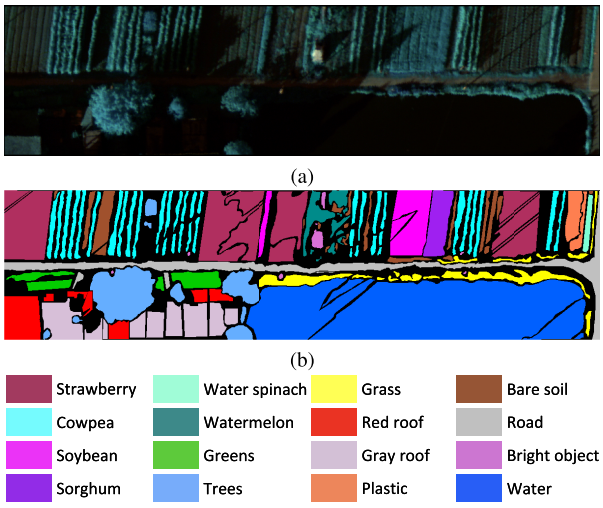



(a) (b)

Fig. 6. Xuzhou dataset. (a) False-color map. (b) Ground-truth map.

TABLE III
TRAIN AND TEST SAMPLES IN THE XUZHOU DATASET

| ID | Land Cover Class | Train | Test |
|---|---|---|---|
| C01 | Bareland_1 | 132 | 26264 |
| C02 | Lakes | 20 | 4007 |
| C03 | Coals | 14 | 2769 |
| C04 | Cement | 26 | 5188 |
| C05 | Crop_1 | 66 | 13118 |
| C06 | Trees | 12 | 2424 |
| C07 | Bareland_2 | 35 | 6955 |
| C08 | Crop_2 | 24 | 4753 |
| C09 | Red_tiles | 15 | 3055 |
| - | Total | 344 | 68533 |




(a)

(b)

Fig. 7. WHU-Hi-HanChuan dataset. (a) False-color map. (b) Ground-truth map.

## B. Experimental Setup

*1) Configurations:* To ensure fair comparisons, both the proposed method and all the comparative methods were implemented using the PyTorch framework. Training and testing were conducted on a server equipped with an Intel Xeon Silver 4210 2.20-GHz CPU and an NVIDIA GeForce RTX 2080Ti GPU with 11 GB of memory. For our method, we employed the Adam optimizer to update the network parameters, with a learning rate of 5e-4 and a batch size of 32; the training epochs are 100 epochs. As for the comparative methods, all settings remained consistent with their original configurations to achieve optimal performance.

TABLE IV
TRAIN AND TEST SAMPLES IN THE WHU-HI-HANCHUAN DATASET

| ID | Land Cover Class | Train | Test |
|---|---|---|---|
| C01 | Strawberry | 447 | 44288 |
| C02 | Cowpea | 227 | 22526 |
| C03 | Soybean | 103 | 10184 |
| C04 | Sorghum | 54 | 5299 |
| C05 | Water spinach | 12 | 1188 |
| C06 | Watermelon | 45 | 4488 |
| C07 | Greens | 59 | 5844 |
| C08 | Trees | 180 | 17798 |
| C09 | Grass | 95 | 9374 |
| C10 | Red roof | 105 | 10411 |
| C11 | Gray roof | 169 | 16742 |
| C12 | Plastic | 37 | 3642 |
| C13 | Bare soil | 91 | 9025 |
| C14 | Road | 186 | 18374 |
| C15 | Bright object | 11 | 1125 |
| C16 | Water | 754 | 74647 |
| - | Total | 2575 | 254955 |

*2) Evaluation Indicators:* We performed a quantitative evaluation of the classification performance of all methods using four well-established evaluation metrics. Of these indicators, single-class accuracy represents the ratio of the number of correctly classified samples for each land cover class to the total number of samples in that class. The overall accuracy (OA) is the ratio of correctly classified samples to the total number of samples across all classes. The average accuracy (AA) is the mean of all single-class accuracies. The kappa coefficient ($\kappa$) serves as a measure of the consistency between the model's classification results and the ground truth. In all these evaluation indicators, higher values indicate better performance.

## C. Parameter Analysis

*1) Remained Spectral Dimensions:* While keeping all other parameters constant, we investigate the impact of the remaining spectral dimensions on the performance of classification by considering a candidate set of values: 10, 20, 30, 40, 50. In addition, we also considered the classification results without performing PCA dimensionality reduction. Fig. 8(a) illustrates the OA of the model across various spectral dimensions for the four datasets. It is evident that without feature dimensionality reduction, the model shows the poorest performance across all four datasets. After feature reduction, among the considered dimension candidate values, with the increase in spectral dimensions, the OA of the PU dataset gradually decreases, while that of the WHU-Hi-HanChuan dataset increases. For the Salinas and Xuzhou datasets, the OA initially rises before stabilizing. Considering both classification performance and computational complexity, we set the remained spectral dimensions to 30.

*2) Patch Size:* To investigate the impact of patch size on classification performance, we maintained other parameters constant and considered a set of candidate values 9, 11, 13, 15, 17 for the patch size. Fig. 8(b) depicts the model's OA across various patch sizes for the four datasets. It is obvious that the PU dataset attains optimal performance with a patch size of 13,

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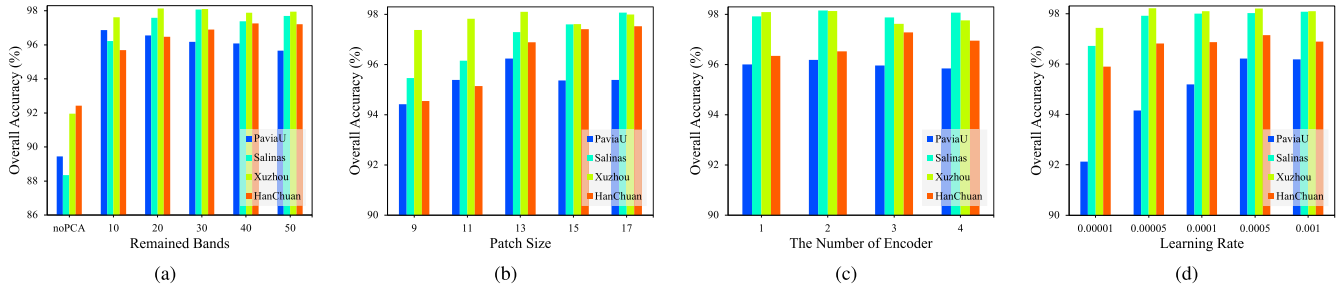



Fig. 8. Impact of different parameters on OA. (a) Remained bands. (b) Patch size. (c) Number of encoders. (d) Learning rate.

TABLE V

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE PU DATASET WITH 0.5% TRAINING SAMPLES (BEST RESULTS ARE BOLDED)

| NO. | CNN-based | | | | | Tansformer-based | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 3D-CNN[27] | M3D-DCNN[28] | DFFN[61] | HybridSN[37] | RSSAN[62] | SF[51] | SSFTT[52] | GAHT[55] | morphFormer[56] | Proposed |
| 1 | 95.77±01.06 | 97.24±01.02 | 96.39±01.73 | 91.32±04.21 | 96.83±01.99 | 87.09±02.15 | 96.24±01.44 | 97.92±01.25 | 93.88±02.23 | **99.00±00.54** |
| 2 | 99.29±00.37 | 98.86±00.34 | 99.19±00.64 | 98.86±00.82 | 99.37±00.28 | 96.51±01.32 | **99.86±00.17** | 99.40±00.47 | 98.69±01.09 | 99.84±00.17 |
| 3 | 35.82±06.77 | 80.99±05.02 | 57.64±21.08 | 83.32±07.22 | 81.30±03.74 | 62.84±05.01 | 76.44±04.58 | 77.11±03.33 | 67.71±12.54 | **83.72±04.42** |
| 4 | 92.14±02.50 | 91.94±02.91 | 89.10±03.06 | 80.00±05.08 | 88.89±02.47 | 84.00±03.04 | 83.28±03.08 | **95.06±01.21** | 93.24±02.14 | 83.65±02.10 |
| 5 | 98.96±00.96 | 99.05±00.97 | 97.82±04.43 | 99.63±00.39 | 90.56±05.11 | 99.80±00.28 | **100.00±00.00** | 99.83±00.14 | 98.29±01.39 | **100.00±00.00** |
| 6 | 34.75±07.40 | 72.43±04.86 | 86.11±06.57 | 94.51±06.53 | 91.69±04.47 | 61.72±05.30 | 94.49±03.75 | 94.97±01.75 | 94.90±04.91 | **97.18±01.36** |
| 7 | 40.96±07.48 | 70.54±06.18 | 60.40±10.23 | 90.56±09.76 | 65.13±09.72 | 54.33±06.29 | **99.51±01.03** | 67.44±07.62 | 75.79±08.55 | 98.84±01.26 |
| 8 | 76.41±01.97 | 78.82±04.41 | 79.11±04.26 | 72.22±07.52 | 76.02±03.60 | 55.07±03.27 | 84.21±04.84 | 80.38±06.28 | **91.53±01.97** | 88.33±03.14 |
| 9 | 98.31±00.69 | 98.74±00.79 | 99.26±00.79 | 56.39±07.02 | 96.55±02.28 | 95.40±01.43 | 86.60±05.79 | **99.80±00.17** | 94.74±02.79 | 89.76±03.47 |
| OA(%) | 83.71±00.95 | 91.53±00.66 | 91.48±01.10 | 91.60±01.28 | 93.02±00.85 | 83.61±00.62 | 94.68±00.81 | 94.64±00.62 | 94.13±01.28 | **96.21±00.56** |
| AA(%) | 74.71±01.17 | 87.62±01.07 | 85.00±02.15 | 85.30±01.48 | 87.37±01.66 | 77.42±00.99 | 91.18±01.39 | 90.21±01.17 | 89.75±02.28 | **93.37±00.95** |
| $\kappa \times 100$ | 77.56±01.40 | 88.59±00.91 | 88.58±01.49 | 88.79±01.71 | 90.67±01.15 | 77.88±00.86 | 92.91±01.09 | 92.86±0.83 | 92.21±01.70 | **94.95±00.75** |

whereas the Salinas and WHU-Hi-HanChuan datasets exhibit an increase in OA with an increment in patch size. As for the Xuzhou dataset, OA initially rises before stabilizing as the patch size increases. Therefore, we set the patch size of the Salinas to 17 and the patch size of the remaining three datasets to 13.

*3) Number of Encoder:* The number of encoders plays a crucial role in the representational capacity of features. We considered a set of values 1, 2, 3, 4 to explore its impact on classification performance. As illustrated in Fig. 8(c), optimal performance is achieved for the PU, Salinas, and Xuzhou datasets when the number of encoders is set to 2, while the WHU-Hi-HanChuan dataset attains optimal performance with a quantity of 3 encoders. Therefore, we set the number of encoders to 2.

*4) Learning Rate:* The learning rate is a pivotal parameter during the model training process, directly influencing the convergence speed of the model. We considered a set of candidate values 1e-5, 5e-5, 1e-4, 5e-4, 1e-3 for the learning rate. Fig. 8(d) depicts the impact of different learning rates on OA. It is clear that with an increase in the learning rate, the classification performance for all four datasets exhibits an initial improvement before reaching stability. Consequently, we set the learning rate to 5e-4.

### D. Classification Results and Analysis

To substantiate the efficacy of the proposed classification method, we conducted comparisons with several representative approaches based on both CNN and transformer architectures. Among the CNN-based methods are 3D-CNN [27], M3D-DCNN [28], DFFN [61], HybridSN [37], and RSSAN [62]; transformer-based methods include SF [51], SSFTT [52], GAHT [55], and morphFormer [56].

*1) Quantitative Results and Analysis:* The single-class accuracy, OA, AA, and $\kappa$ for the proposed method and all comparative methods on the four datasets are shown in Tables V–VIII, where the best results are highlighted in bold. In Table V, it can be observed that although CNN-based methods have achieved good classification accuracy, the lack of modeling global dependencies resulted in all best single-class accuracies appearing in transformer-based methods. In addition, our proposed method achieved the highest number of best single-class accuracies, further validating the superiority of our proposed approach. However, our proposed method also has some limitations. For instance, the accuracy achieved on the Bitumen class with imbalanced sample size is slightly inferior to SSFTT. This might be because SSFTT excels at learning high-level semantic features. From Table VI, it is evident that several small sample classes in the Salinas dataset, such as Lettuce romaine 6wk and Lettuce romaine 7wk, showed good performance with methods based on CNN. This could be due to their ability to extract local detailed information effectively. As shown in Table VII, the performance of SF was inferior to that of 3D-CNN, possibly because the Xuzhou dataset has the most spectral dimensions. SF tends to directly capture long-distance dependencies across the complete spectrum, resulting in a large amount of redundant information and consequently poorer model performance. In Table VIII, despite category 11 gray roof and category 14 road having similar spectral responses, the proposed method excels in distinguishing between these land cover types. Overall, our proposed method

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TABLE VI

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE SALINAS DATASET WITH 0.5% TRAINING SAMPLES (BEST RESULTS ARE BOLDED)

| NO. | CNN-based | | | | | Tansformer-based | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 3D-CNN[27] | M3D-DCNN[28] | DFFN[61] | HybridSN[37] | RSSAN[62] | SF[51] | SSFTT[52] | GAHT[55] | morphFormer[56] | Proposed |
| 1 | 96.13±02.10 | 96.81±02.42 | 96.77±03.13 | 99.62±00.79 | 97.29±01.75 | 93.14±01.98 | 99.92±00.16 | 99.56±00.89 | 97.17±05.17 | **100.00±00.00** |
| 2 | 99.69±00.41 | 99.83±00.38 | 98.58±02.06 | 99.99±00.03 | 98.88±01.08 | 98.75±00.62 | **100.00±00.00** | 99.72±00.60 | 99.17±02.09 | **100.00±00.00** |
| 3 | 88.35±05.34 | 96.67±00.75 | 88.82±05.24 | 98.75±01.04 | 95.92±02.65 | 93.57±02.74 | 99.92±00.19 | 97.16±02.24 | 93.44±07.90 | **100.00±00.00** |
| 4 | 97.31±00.93 | 95.38±01.88 | 94.78±01.38 | 81.12±08.19 | 95.60±03.92 | 90.39±01.48 | 97.92±02.39 | **98.32±01.40** | 94.05±03.08 | 96.87±01.03 |
| 5 | 94.65±01.51 | 96.94±00.59 | 92.38±02.68 | 98.62±02.55 | 95.47±04.46 | 81.20±01.61 | 99.93±00.13 | 97.10±02.32 | 93.80±04.13 | **100.00±00.00** |
| 6 | 99.69±00.22 | 99.97±00.03 | 99.74±00.28 | 98.96±01.14 | 99.67±00.49 | 99.90±00.17 | 99.33±00.78 | **99.99±00.03** | 98.44±02.47 | 98.95±00.72 |
| 7 | 99.41±00.14 | 99.67±00.17 | 99.12±00.42 | 98.26±03.63 | 97.90±01.20 | 98.69±00.39 | 99.21±01.13 | 99.73±00.30 | 98.90±01.03 | **99.89±00.12** |
| 8 | 75.61±03.06 | 85.52±02.15 | 84.93±03.67 | 95.81±04.36 | 83.17±04.61 | 80.56±02.53 | 93.02±01.58 | 88.86±03.46 | 93.18±02.35 | **97.90±01.69** |
| 9 | 95.96±00.38 | 96.81±01.35 | 95.30±00.93 | **100.00±00.00** | 98.96±01.20 | 94.67±00.95 | **100.00±00.00** | 98.62±01.45 | 99.58±00.50 | **100.00±00.00** |
| 10 | 89.81±01.07 | 94.91±00.85 | 90.04±03.22 | 98.87±01.44 | 95.54±00.82 | 88.43±01.36 | 98.64±00.81 | 98.46±01.33 | 96.49±03.92 | **99.76±00.25** |
| 11 | 62.87±03.08 | 77.62±04.55 | 62.11±06.51 | 87.69±06.23 | 75.38±04.03 | 75.72±05.62 | 99.54±00.90 | 91.43±06.62 | 95.63±05.63 | **99.70±00.12** |
| 12 | 99.06±00.30 | 99.14±00.58 | 97.55±04.51 | 98.00±02.47 | 96.77±02.46 | 97.37±01.17 | 99.31±00.89 | 99.37±00.85 | 95.15±06.72 | **99.92±00.07** |
| 13 | 99.65±00.15 | **99.69±00.19** | 89.27±17.64 | 85.18±11.52 | 97.62±01.92 | 96.48±02.50 | 90.77±06.62 | 93.13±07.02 | 94.91±01.86 | 88.34±05.04 |
| 14 | 96.22±01.08 | 95.87±01.25 | **96.70±01.78** | 95.04±04.53 | 92.40±02.87 | 92.71±01.65 | 94.49±02.53 | 93.11±06.86 | 95.02±03.38 | 93.62±01.19 |
| 15 | 69.73±04.38 | 73.87±03.83 | 74.08±05.64 | **94.80±03.02** | 84.66±05.84 | 77.63±02.50 | 88.78±02.04 | 90.42±05.92 | 87.62±03.93 | 91.59±00.98 |
| 16 | 74.48±00.79 | 73.83±00.90 | 77.54±03.63 | 97.79±06.31 | 86.71±04.19 | 74.77±01.88 | 98.39±02.30 | 86.32±05.98 | 98.39±02.16 | **99.99±00.03** |
| OA(%) | 87.09±00.26 | 90.82±00.21 | 89.19±01.30 | 96.82±00.96 | 91.98±00.39 | 88.17±00.52 | 96.44±00.24 | 94.90±00.74 | 95.11±00.68 | **97.92±00.18** |
| AA(%) | 89.91±00.40 | 92.66±00.33 | 89.86±01.52 | 95.53±01.01 | 93.25±00.97 | 89.62±00.44 | 97.45±00.42 | 95.71±00.56 | 95.69±00.57 | **97.91±00.23** |
| $\kappa \times 100$ | 85.64±00.29 | 89.78±00.23 | 87.97±01.45 | 96.46±01.07 | 91.09±00.44 | 86.85±00.57 | 96.04±00.26 | 94.32±00.83 | 94.55±00.76 | **97.68±00.20** |

TABLE VII

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE XUZHOU DATASET WITH 0.5% TRAINING SAMPLES (BEST RESULTS ARE BOLDED)

| NO. | CNN-based | | | | | Tansformer-based | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 3D-CNN[27] | M3D-DCNN[28] | DFFN[61] | HybridSN[37] | RSSAN[62] | SF[51] | SSFTT[52] | GAHT[55] | morphFormer[56] | Proposed |
| 1 | 96.98±00.60 | 98.14±00.38 | 97.75±00.60 | 96.62±00.77 | 97.92±00.51 | 96.90±00.62 | 96.72±00.65 | **98.36±00.65** | 97.08±01.00 | 98.11±00.31 |
| 2 | 96.13±00.88 | 97.64±00.59 | 96.91±01.08 | 94.86±03.98 | 98.45±00.54 | 95.96±00.50 | 98.05±00.80 | **99.69±00.15** | 98.51±00.86 | 98.60±00.71 |
| 3 | 77.12±04.33 | 81.42±01.86 | 78.39±02.34 | 83.29±04.47 | 81.45±04.01 | 78.90±01.83 | **95.08±01.59** | 85.48±07.19 | 89.22±03.77 | 93.13±03.22 |
| 4 | 91.80±01.34 | 92.88±00.98 | 91.31±01.35 | 95.02±02.33 | 94.94±01.91 | 86.31±02.28 | 99.38±00.21 | 94.12±01.22 | 94.85±01.49 | **99.44±00.31** |
| 5 | 96.37±00.70 | 97.19±00.64 | 96.01±00.83 | 97.40±01.40 | 97.24±00.66 | 93.91±00.98 | **98.98±00.35** | 97.22±00.56 | 98.28±00.94 | 98.56±00.19 |
| 6 | 75.69±04.21 | 84.36±04.33 | 82.62±05.55 | 96.50±03.03 | 83.64±04.81 | 82.54±02.27 | 97.73±01.73 | 91.83±06.13 | 89.93±03.72 | **99.10±00.50** |
| 7 | 93.25±02.91 | 89.53±02.43 | 94.49±02.51 | 97.84±01.19 | 93.92±02.85 | 88.89±02.24 | **99.25±00.33** | 93.72±04.21 | 94.61±01.52 | 98.82±00.33 |
| 8 | 97.14±00.87 | 97.96±00.76 | 97.71±00.71 | 98.83±03.07 | 98.47±00.78 | 95.17±00.88 | 99.49±00.46 | 97.11±01.07 | 97.47±01.19 | **99.50±00.19** |
| 9 | 90.33±04.58 | 95.89±02.77 | 91.14±05.06 | 90.46±07.52 | **97.33±01.87** | 93.97±01.58 | 93.98±03.09 | 95.45±02.54 | 95.26±02.47 | 96.41±00.81 |
| OA(%) | 94.20±00.34 | 95.38±00.25 | 94.93±00.70 | 96.01±00.44 | 96.03±00.39 | 93.17±00.43 | 97.73±00.30 | 96.46±00.70 | 96.35±00.54 | **98.25±00.24** |
| AA(%) | 90.54±00.71 | 92.78±00.64 | 91.81±01.27 | 94.53±01.25 | 93.71±00.72 | 90.27±00.70 | 97.63±00.35 | 94.78±01.55 | 95.02±00.97 | **97.96±00.39** |
| $\kappa \times 100$ | 92.63±00.43 | 94.12±00.33 | 93.56±00.89 | 94.94±00.56 | 94.96±00.50 | 91.29±00.55 | 97.12±00.37 | 95.50±00.89 | 95.37±00.68 | **97.78±00.31** |

TABLE VIII

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE WHU-HI-HANCHUAN DATASET WITH 1% TRAINING SAMPLES (BEST RESULTS ARE BOLDED)

| NO. | CNN-based | | | | | Tansformer-based | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 3D-CNN[27] | M3D-DCNN[28] | DFFN[61] | HybridSN[37] | RSSAN[62] | SF[51] | SSFTT[52] | GAHT[55] | morphFormer[56] | Proposed |
| 1 | 95.19±00.45 | 95.84±00.84 | 96.75±00.63 | **99.26±00.47** | 96.38±00.95 | 94.40±00.63 | 98.49±00.89 | 97.90±00.80 | 98.71±00.42 | 99.01±00.26 |
| 2 | 87.84±01.13 | 89.37±02.81 | 93.76±01.29 | 96.36±02.02 | 93.53±01.68 | 83.17±02.66 | 94.06±01.16 | 95.84±01.18 | 95.96±01.02 | **96.50±00.96** |
| 3 | 84.48±02.01 | 91.22±02.56 | 94.36±01.66 | 97.16±01.46 | 92.45±01.77 | 94.51±01.29 | 93.57±04.83 | **98.08±00.73** | 95.70±01.94 | 97.38±02.58 |
| 4 | 91.21±01.84 | 94.65±00.78 | 96.28±00.74 | **99.67±00.32** | 96.75±00.90 | 95.66±00.59 | 98.84±00.55 | 98.55±00.62 | 98.82±00.41 | 95.19±06.84 |
| 5 | 52.01±04.85 | 62.83±11.82 | 84.20±04.51 | **94.71±05.04** | 77.04±05.73 | 67.98±07.23 | 89.52±05.81 | 85.01±03.76 | 91.66±04.79 | 90.66±05.84 |
| 6 | 24.85±02.86 | 55.13±09.61 | 62.09±08.03 | 70.55±05.83 | 60.89±07.50 | 42.64±02.40 | 73.08±07.80 | 74.47±01.90 | 69.51±05.33 | **83.91±02.09** |
| 7 | 84.39±03.08 | 82.66±07.40 | 88.28±01.96 | **93.35±03.04** | 84.90±08.64 | 79.41±05.76 | 92.63±01.17 | 91.93±03.01 | 92.29±02.00 | 92.09±01.14 |
| 8 | 77.17±01.55 | 81.23±03.99 | 87.45±03.44 | 92.32±02.43 | 86.33±03.52 | 80.72±02.31 | **94.17±01.58** | 90.62±02.56 | 91.96±01.90 | 91.84±07.63 |
| 9 | 73.30±04.82 | 79.50±05.69 | 90.18±04.67 | 90.70±02.46 | 85.57±02.67 | 71.28±02.57 | **95.42±01.81** | 91.39±03.52 | 92.49±02.11 | 93.01±11.60 |
| 10 | 90.42±00.81 | 95.12±01.77 | 97.35±01.15 | 97.94±01.44 | 96.51±01.42 | 95.18±00.74 | 98.14±01.10 | **98.91±00.47** | 98.80±00.56 | 98.69±00.75 |
| 11 | 93.98±01.15 | 95.81±02.41 | 95.70±00.39 | 96.53±01.75 | 95.88±01.66 | 93.47±01.23 | 97.85±00.51 | 96.49±01.11 | 97.86±00.73 | **98.44±00.56** |
| 12 | 30.43±01.96 | 55.14±11.83 | 76.29±04.26 | 85.85±10.01 | 71.19±07.58 | 46.84±07.28 | 87.41±04.91 | 83.91±03.12 | 89.21±04.74 | **93.51±04.76** |
| 13 | 62.41±01.96 | 68.32±05.60 | 79.54±01.26 | 81.25±02.70 | 77.20±04.05 | 70.87±03.86 | 79.77±03.00 | **84.79±02.18** | 81.84±01.39 | 80.65±05.99 |
| 14 | 85.93±00.98 | 88.96±03.12 | 91.72±01.21 | 94.31±02.16 | 90.17±04.28 | 86.27±01.49 | 95.30±01.07 | 94.19±01.63 | 95.41±01.25 | **96.03±00.72** |
| 15 | 79.31±01.72 | 75.20±02.96 | 75.20±06.20 | 41.33±10.82 | 69.89±07.75 | 64.25±06.06 | 80.13±08.73 | 82.10±03.21 | **84.50±06.39** | 82.87±02.78 |
| 16 | 99.36±00.09 | 99.56±00.19 | 99.56±00.15 | 99.83±00.23 | 99.50±00.31 | 99.29±00.20 | 99.48±00.18 | 99.75±00.10 | **99.85±00.09** | 99.79±00.08 |
| OA(%) | 88.41±00.44 | 91.18±00.64 | 94.05±00.62 | 95.96±00.53 | 93.18±00.67 | 89.43±00.37 | 95.97±00.30 | 95.87±00.40 | 96.20±00.28 | **96.62±01.05** |
| AA(%) | 75.77±00.97 | 81.91±01.59 | 88.04±01.37 | 89.45±00.93 | 85.89±01.57 | 79.12±00.97 | 91.74±00.76 | 91.50±00.65 | 92.16±00.91 | **93.10±01.89** |
| $\kappa \times 100$ | 86.40±00.52 | 89.66±00.75 | 93.03±00.73 | 95.27±00.63 | 92.02±00.78 | 87.62±00.44 | 95.29±00.35 | 95.17±00.47 | 95.55±00.32 | **96.05±01.23** |

outperforms other approaches in terms of classification performance across all datasets. Taking OA as an example, MASSFormer achieves higher accuracy than the best accuracy among the comparative methods by 1.53%, 1.1%, 0.52%, and 0.42% on the PU, Salinas, Xuzhou, and WHU-Hi-HanChuan datasets, respectively.

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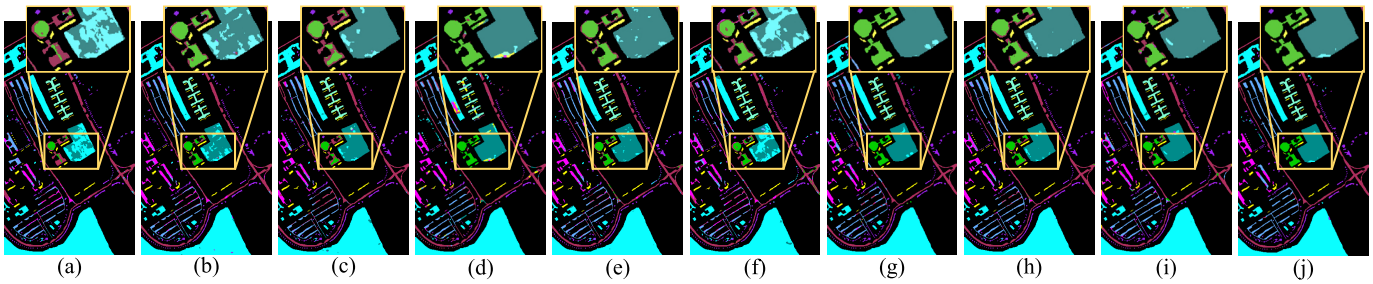



Fig. 9. Classification maps of the PU dataset: (a) 3D-CNN, (b) M3D-DCNN, (c) DFFN, (d) HybridSN, (e) RSSAN, (f) SF, (g) SSFTT, (h) GAHT, (i) morphFormer, and (j) proposed.

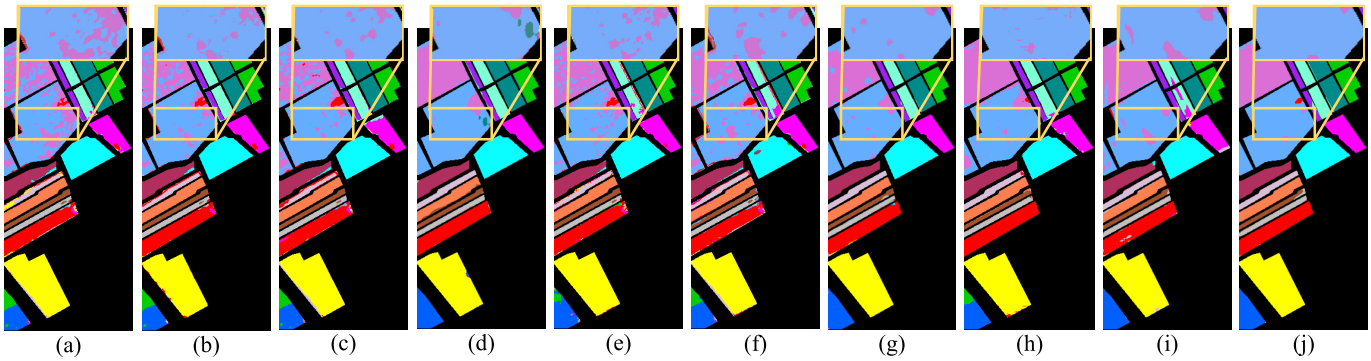



Fig. 10. Classification maps of the Salinas dataset: (a) 3D-CNN, (b) M3D-DCNN, (c) DFFN, (d) HybridSN, (e) RSSAN, (f) SF, (g) SSFTT, (h) GAHT, (i) morphFormer, and (j) proposed.

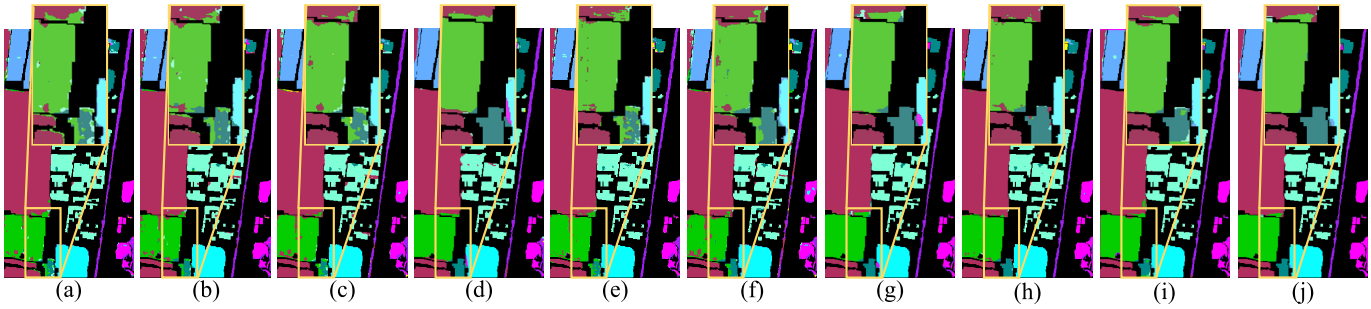



Fig. 11. Classification maps of the Xuzhou dataset (a) 3D-CNN, (b) M3D-DCNN, (c) DFFN, (d) HybridSN, (e) RSSAN, (f) SF, (g) SSFTT, (h) GAHT, (i) morphFormer, and (j) proposed.

*2) Visual Evaluation and Analysis:* Figs. 9–12 illustrate the classification maps of the proposed method and the comparative methods for all four datasets. It is clear that the classification maps generated by MASSFormer show reduced noise and achieve a closer overall performance to the ground truth. Taking the Salinas dataset as an example, our method demonstrates significantly superior classification performance for the Grapes_untrained and Vinyard_untrained land cover classes compared to other classification methods. SSFTT and HybridSN closely follow, also achieving relatively good classification results.

*3) Efficiency Evaluation:* As illustrated in Fig. 13, we conducted a detailed comparison of the training parameters and performance between the proposed method and the comparative methods. In the graph, the horizontal and vertical axes represent OA and AA, respectively, while the circle's radius indicates the number of parameters. It is evident from the figure that MASSFormer offers superior classification performance with relatively lower computational complexity.

TABLE IX

RUNTIME OF ALL METHODS ON FOUR DATASETS (OPTIMAL RESULTS ARE BOLDED AND SUBOPTIMAL RESULTS ARE UNDERLINED)

| Methods | Pavia University | | Salinas | | Xuzhou | | WHU-Hi-HanChuan | |
|---|---|---|---|---|---|---|---|---|
| | Train(m) | Test(s) | Train(m) | Test(s) | Train(m) | Test(s) | Train(m) | Test(s) |
| 3D-CNN[27] | 0.33 | 5.59 | 0.56 | 4.93 | 1.21 | 11.65 | 6.40 | 22.79 |
| M3D-DCNN[28] | 0.36 | 6.01 | 0.63 | 5.14 | 1.44 | 12.86 | 7.11 | 24.99 |
| DFFN[61] | 0.88 | 20.36 | 1.02 | 10.09 | 2.00 | 18.39 | 10.76 | 42.73 |
| HybridSN[37] | **0.16** | **2.93** | 0.36 | 9.23 | 0.24 | **5.53** | **1.15** | **21.25** |
| RSSAN[62] | 0.59 | 11.74 | 0.84 | 6.90 | 1.50 | 10.74 | 7.73 | 26.16 |
| SF[51] | 1.10 | 33.73 | 1.61 | 24.52 | 4.44 | 68.78 | 17.55 | 101.21 |
| SSFTT[52] | **0.16** | 3.22 | **0.19** | **4.28** | **0.23** | 5.62 | 1.33 | 22.07 |
| GAHT[55] | 0.77 | 17.47 | 1.14 | 10.17 | 1.78 | 18.16 | 9.26 | 39.92 |
| morphFormer[56] | 0.82 | 12.37 | 1.92 | 26.66 | 1.87 | 28.25 | 8.39 | 78.44 |
| Proposed | 0.23 | 5.35 | 0.30 | 7.65 | 0.37 | 9.36 | 2.38 | 33.84 |

In addition, to assess the efficiency of the proposed method, we recorded the training and testing times for all classification methods, as presented in Table IX. HybridSN and SSFTT demonstrate optimal operational efficiency, closely followed by MASSFormer, which shows training and testing times that are relatively close to the optimal methods. Overall, our

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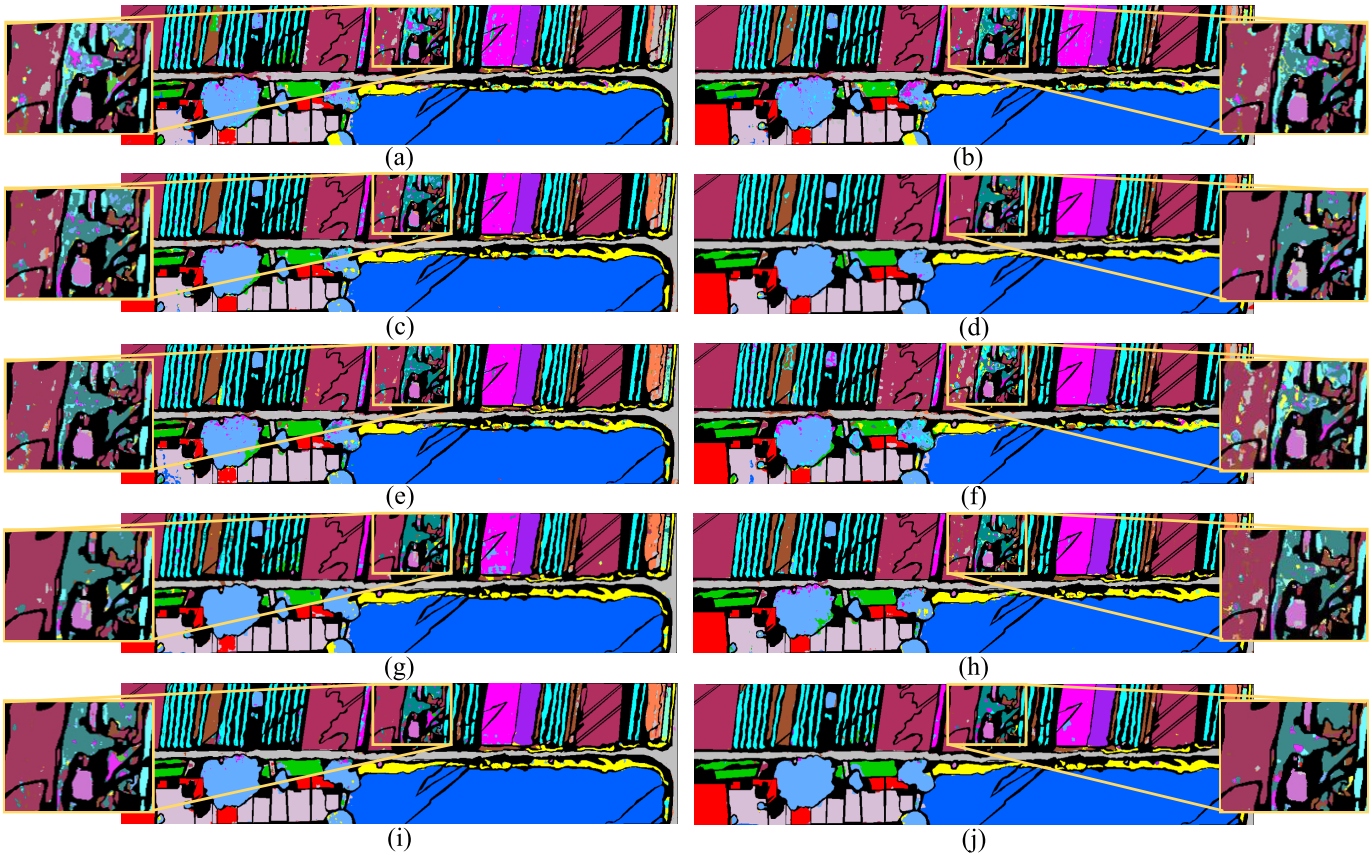



Fig. 12. Classification maps of the WHU-Hi-HanChuan dataset: (a) 3D-CNN, (b) M3D-DCNN, (c) DFFN, (d) HybridSN, (e) RSSAN, (f) SF, (g) SSFTT, (h) GAHT, (i) morphFormer, and (j) proposed.

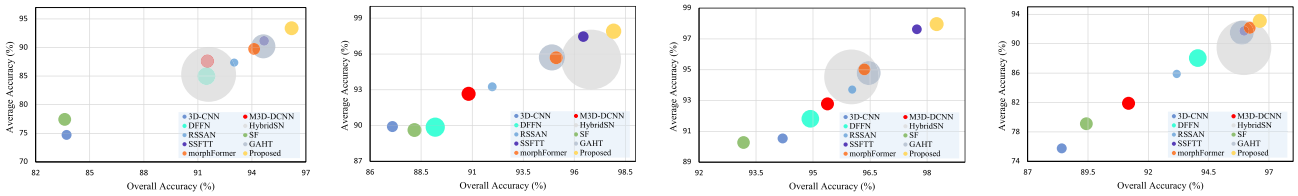



Fig. 13. Comparison of different models in terms of OA, AA, and training parameter count. (a) PU. (b) Salinas. (c) Xuzhou. (d) WHU-Hi-HanChuan.

proposed method not only excels in classification accuracy but also demonstrates satisfactory performance in terms of execution efficiency.

*4) Robustness Evaluation:* To evaluate the robustness of the proposed method, we conducted tests on the classification accuracy of all methods under different sampling rates across the four datasets. For the PU, Salinas, and Xuzhou datasets, we randomly selected 0.25%, 0.5%, 0.75%, and 1% of samples for training. For the WHU-Hi-HanChuan dataset, we randomly selected 0.5%, 1%, 1.5%, and 2% of samples for training. Fig. 14 shows the OA of all methods at different sampling rates. It is observed that, with a very limited number of training samples, some methods experience a significant decrease in classification accuracy, while the proposed method maintains a relatively high accuracy and exhibits strong robustness. Overall, except for the PU dataset with a sampling rate of 0.25%, where our method slightly lags behind GAHT, the proposed method exhibits optimal classification performance in all other scenarios.

## E. Ablation Study

Due to the inclusion of various components in our proposed method, we conducted ablation experiments on the PU dataset to assess the impact of each component on classification performance. The primary components of the proposed method include Conv3D, Conv2D, MT, and MATE. Evaluation metrics include OA, AA, and $\kappa$, and the experimental results are presented in Table X.

Specifically, Case 1 involves the removal of Conv3D and Conv2D, directly feeding the dimension-reduced HSI patches into the remaining modules for training. Cases 2 and 3 retain Conv3D and Conv2D, respectively. The findings reveal that, compared to Cases 2 and 3, Case 1 yields the lowest values for all three indicators, indicating that the absence of convolutional

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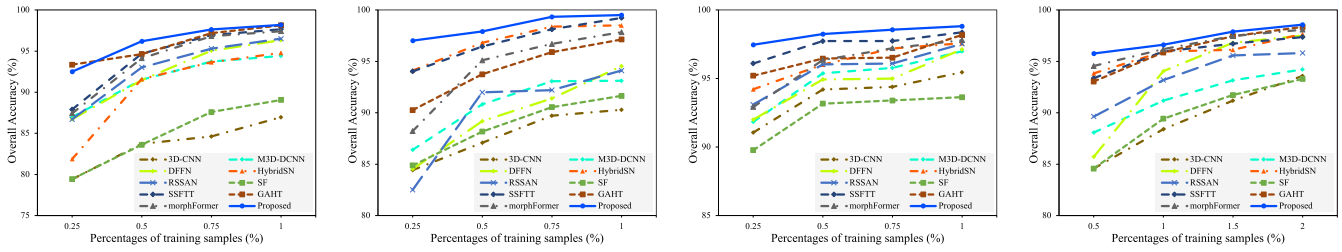



Fig. 14. Impact of different training sample percentages on OA. (a) PU. (b) Salinas. (c) Xuzhou. (d) WHU-Hi-HanChuan.

TABLE X

ABLATION ANALYSIS OF DIFFERENT COMPONENTS IN THE PROPOSED MODEL ON THE PU DATASET (OPTIMAL RESULTS ARE BOLDED)

| Cases | Component | | | | Indicators | | |
|---|---|---|---|---|---|---|---|
| | Conv3D | Conv2D | MT | MATE | OA (%) | AA (%) | $\kappa \times 100$ |
| 1 | - | - | √ | √ | 93.53 | 88.45 | 91.34 |
| 2 | √ | - | √ | √ | 94.86 | 90.84 | 93.15 |
| 3 | - | √ | √ | √ | 94.48 | 90.86 | 92.63 |
| 4 | √ | √ | - | - | 87.20 | 74.59 | 82.65 |
| 5 | √ | √ | √ | √ | **96.21** | **93.37** | **94.95** |

blocks limits the spatial feature representation capability of the data. Furthermore, the results from Cases 2 and 3 demonstrate that Conv3D, performing joint convolution in both spatial and spectral dimensions, outperforms Conv2D, which only convolves in the spatial dimension. Case 4 is the only scenario without the MT and MATE modules, resulting in the poorest performance. This suggests that relying solely on the local feature extraction capability of the convolutional blocks is insufficient, emphasizing the crucial role of deep-level feature learning provided by MT and MATE in improving classification accuracy. Finally, experimental results from Cases 2, 3, and 5 indicate that the hybrid design of Conv3D and Conv2D is effective, balancing classification accuracy while maintaining lower computational complexity. In conclusion, each component of the proposed method plays a positive role in achieving the final classification performance.

## IV. CONCLUSION

In this article, we proposed a MASSFormer method for HSI classification. This approach efficiently combines shallow CNNs and transformers, preserving spatial feature extraction capabilities while considering the long-range dependencies between spectral sequence properties and spatial features. In addition, we designed two efficient modules: MT and MATE. MT transforms spectral–spatial features into memory tokens, serving as a persistent repository preserving more original information. Subsequently, MATE introduces these tokens into the model, mitigating modeling difficulties under insufficient feature information and facilitating the network in learning more representative feature representations. Extensive experiments on four public datasets validated the effectiveness of the proposed method.

Our study possesses certain limitations that we plan to address in subsequent research. First, we noticed that the model's performance does not consistently improve with an increase in the number of memory tokens. Therefore, a potential direction is to explore their capacity to encode multimodal features, aiming for a more comprehensive and rich information representation to support the model's tasks. In addition, as the model's depth increases, the advantages conferred by memory tokens appear to plateau. Hence, we intend to investigate the potential of incrementally expanding or updating memory tokens as the model evolves.

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